



Unit 6: The Costs of Production

Instructor: Nguyen Tai Vuong
School of Economics and Management
Hanoi University of Science and Technology



ACTIVE LEARNING 1 Brainstorming costs

You run electric bike.

- List 3 different costs you have.
- List 3 different business decisions that are affected by your costs.



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Objectives

In this unit, look for the answers to these questions:

- What is a production function? What is marginal product? How are they related?
- What are the various costs, and how are they related to each other and to output?
- How are costs different in the short run vs. the long run?
- What are “economies of scale”?

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- 6.1 Total Revenue, Total Cost, Profit
- 6.2 Production Function
- 6.3 Variety of Costs
- 6.4 Costs in the Short Run & Long Run

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6.1 Total Revenue, Total Cost, Profit

- We assume that the firm's goal is to maximize profit.

$$\text{Profit} = \text{Total revenue} - \text{Total cost}$$

the amount a
firm receives
from the sale
of its output

the market
value of the
inputs a firm
uses in
production

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Costs: Explicit vs. Implicit

- **Explicit costs** require an outlay of money, e.g., paying wages to workers.
- **Implicit costs** do not require a cash outlay, e.g., the opportunity cost of the owner's time.
- Remember one of the Ten Principles:
The cost of something is what you give up to get it.
- This is true whether the costs are implicit or explicit. Both matter for firms' decisions.

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Explicit vs. Implicit Costs: An Example

**You need \$100,000 to start your business.
The interest rate is 5%/year.**

- Case 1: borrow \$100,000
 - explicit cost = \$5000 interest on loan
- Case 2: use \$40,000 of your savings, borrow the other \$60,000
 - explicit cost = \$3000 (5%) interest on the loan
 - implicit cost = \$2000 (5%) *foregone* interest you could have earned on your \$40,000.

In both cases, total (exp + imp) costs are \$5000 (a year).

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Economic Profit vs. Accounting Profit

- **Accounting profit**
 - = total revenue minus total explicit costs
- **Economic profit**
 - = total revenue minus total costs
(including explicit and implicit costs)
- Accounting profit ignores implicit costs,
so it's higher than economic profit.

Accounting Profit > Economic Profit

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ACTIVE LEARNING 2

Economic profit vs. accounting profit

The equilibrium rent on office space has just increased by \$500/month.

Compare the effects on accounting profit and economic profit if

- a. you **rent** your office space
- b. you **own** your office space



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Answer

	Office space is rented	Office space is your own
Revenue		
- Explicit cost		
Accounting Profit		
- Implicit cost		
Economic profit		



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6.2 The Production Function

- A **production function** shows the relationship between the quantity of inputs used to produce a good and the quantity of output of that good.
- It can be represented by a table, equation, or graph.
- Example 1:
 - Farmer Jack grows wheat.
 - He has 5 acres of land.
 - He can hire as many workers as he wants.

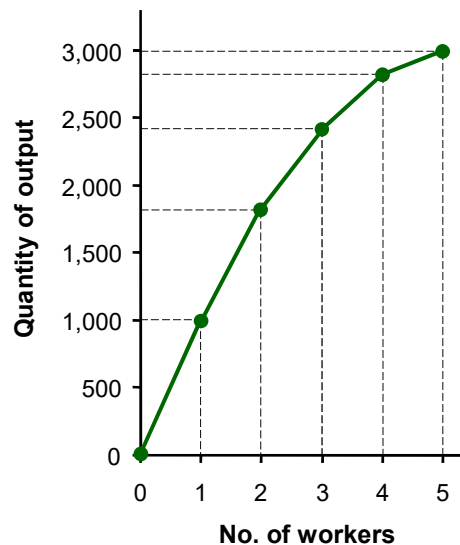


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Example 1: Farmer Jack's Production Function

L (no. of workers)	Q (bushels of wheat)
0	0
1	1000
2	1800
3	2400
4	2800
5	3000



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Marginal Product

- If Jack hires one more worker, his output rises by the *marginal product of labor*.
- The **marginal product** of any input is the increase in output arising from an additional unit of that input, holding all other inputs constant.
- Notation:
 Δ (delta) = "change in..."

Examples:

ΔQ = change in output, ΔL = change in labor

$$\text{Marginal product of labor (MPL)} = \frac{\Delta Q}{\Delta L}$$

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EXAMPLE 1: Total & Marginal Product

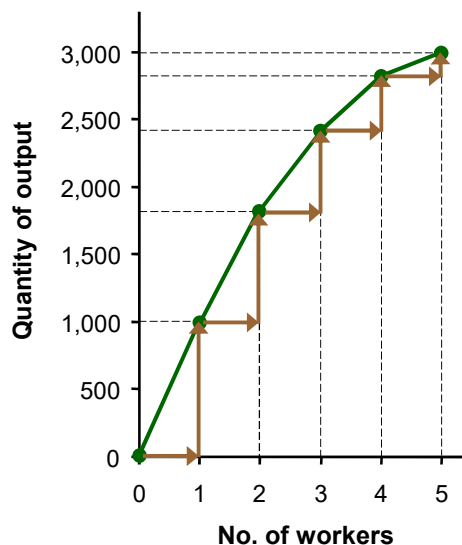
	L (no. of workers)	Q (bushels of wheat)		MPL
	0	0		
$\Delta L = 1$	1	1000	$\Delta Q = 1000$	1000
$\Delta L = 1$	2	1800	$\Delta Q = 800$	800
$\Delta L = 1$	3	2400	$\Delta Q = 600$	600
$\Delta L = 1$	4	2800	$\Delta Q = 400$	400
$\Delta L = 1$	5	3000	$\Delta Q = 200$	200

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EXAMPLE 1: $MPL = \text{Slope of Prod Function}$

L (no. of workers)	Q (bushels of wheat)	MPL
0	0	
1	1000	1000
2	1800	800
3	2400	600
4	2800	400
5	3000	200



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Why MPL Is Important

- Recall one of the Ten Principles:
Rational people think at the margin.
- When Farmer Jack hires an extra worker,
 - his costs rise by the wage he pays the worker
 - his output rises by MPL
- Comparing them helps Jack decide whether he would benefit from hiring the worker.

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Why MPL Diminishes

- Farmer Jack's output rises by a smaller and smaller amount for each additional worker. Why?
- As Jack adds workers, the average worker has less land to work with and will be less productive.
- In general, *MPL* diminishes as *L* rises whether the fixed input is land or capital (equipment, machines, etc.).
- **Diminishing marginal product:** the marginal product of an input declines as the quantity of the input increases (other things equal)

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EXAMPLE 1: Farmer Jack's Costs

- Farmer Jack must pay \$1000 per month for the land, regardless of how much wheat he grows.
- The market wage for a farm worker is \$2000 per month.
- So Farmer Jack's costs are related to how much wheat he produces....

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EXAMPLE 1: Farmer Jack's Costs

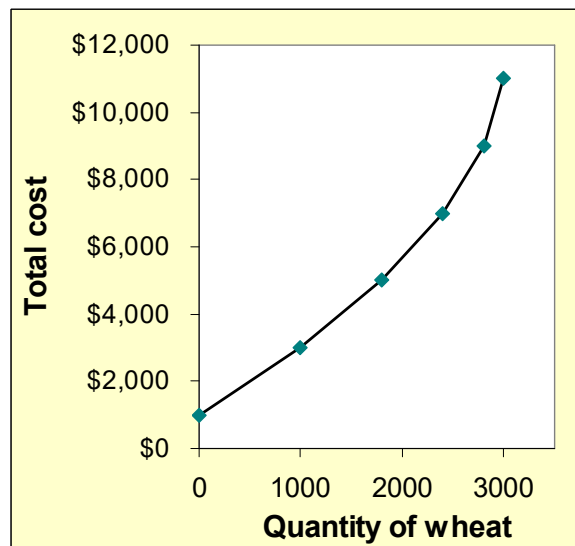
L (no. of workers)	Q (bushels of wheat)	Cost of land	Cost of labor	Total Cost
0	0	\$1,000	\$0	\$1,000
1	1000	\$1,000	\$2,000	\$3,000
2	1800	\$1,000	\$4,000	\$5,000
3	2400	\$1,000	\$6,000	\$7,000
4	2800	\$1,000	\$8,000	\$9,000
5	3000	\$1,000	\$10,000	\$11,000

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EXAMPLE 1: Farmer Jack's Total Cost Curve

Q (bushels of wheat)	Total Cost
0	\$1,000
1000	\$3,000
1800	\$5,000
2400	\$7,000
2800	\$9,000
3000	\$11,000



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6.3 Variety of Costs

- **Marginal Cost (MC)**
is the increase in Total Cost from producing one more unit:

$$MC = \frac{\Delta TC}{\Delta Q}$$

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EXAMPLE 1: Total and Marginal Cost

$$MC = \frac{\Delta TC}{\Delta Q}$$

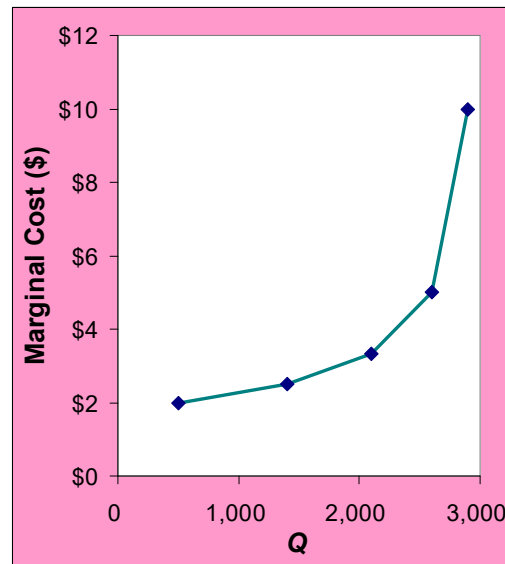
	Q (bushels of wheat)	Total Cost		Marginal Cost (MC)
	0	\$1,000		
$\Delta Q = 1000$	1000	\$3,000	$\Delta TC = \$2000$	\$2.00
$\Delta Q = 800$	1800	\$5,000	$\Delta TC = \$2000$	\$2.50
$\Delta Q = 600$	2400	\$7,000	$\Delta TC = \$2000$	\$3.33
$\Delta Q = 400$	2800	\$9,000	$\Delta TC = \$2000$	\$5.00
$\Delta Q = 200$	3000	\$11,000	$\Delta TC = \$2000$	\$10.00

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EXAMPLE 1: The Marginal Cost Curve

Q (bushels of wheat)	TC	MC
0	\$1,000	
1000	\$3,000	\$2.00
1800	\$5,000	\$2.50
2400	\$7,000	\$3.33
2800	\$9,000	\$5.00
3000	\$11,000	\$10.00



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Why MC Is Important

- Farmer Jack is rational and wants to maximize his profit. To increase profit, should he produce more or less wheat?
- To find the answer, Farmer Jack needs to “think at the margin.”
- If the cost of additional wheat (MC) is less than the revenue he would get from selling it, then Jack’s profits rise if he produces more.

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Fixed and Variable Costs

- **Fixed costs (FC)** do not vary with the quantity of output produced.
 - For Farmer Jack, $FC = \$1000$ for his land
 - Other examples:
cost of equipment, loan payments, rent
- **Variable costs (VC)** vary with the quantity produced.
 - For Farmer Jack, $VC =$ wages he pays workers
 - Other example: cost of materials
- **Total cost (TC)** = $FC + VC$

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EXAMPLE 2

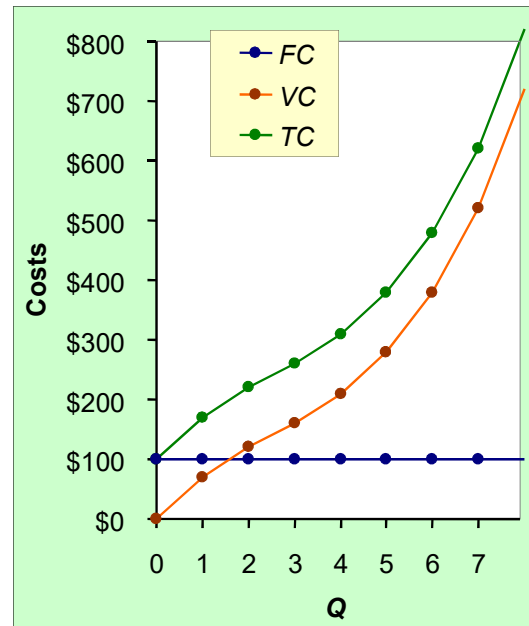
- Our second example is more general,
applies to any type of firm
producing any good with any types of inputs.

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EXAMPLE 2: Costs

Q	FC	VC	TC
0	\$100	\$0	\$100
1	100	70	170
2	100	120	220
3	100	160	260
4	100	210	310
5	100	280	380
6	100	380	480
7	100	520	620

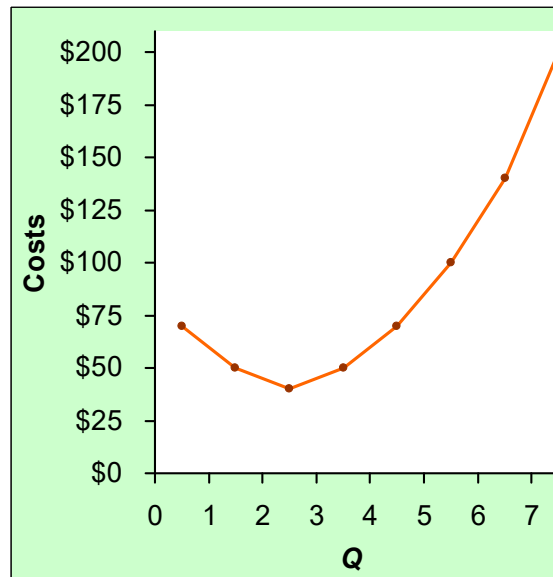


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EXAMPLE 2: Marginal Cost

Q	TC	MC
0	\$100	
1	170	\$70
2	220	50
3	260	40
4	310	50
5	380	70
6	480	100
7	620	140

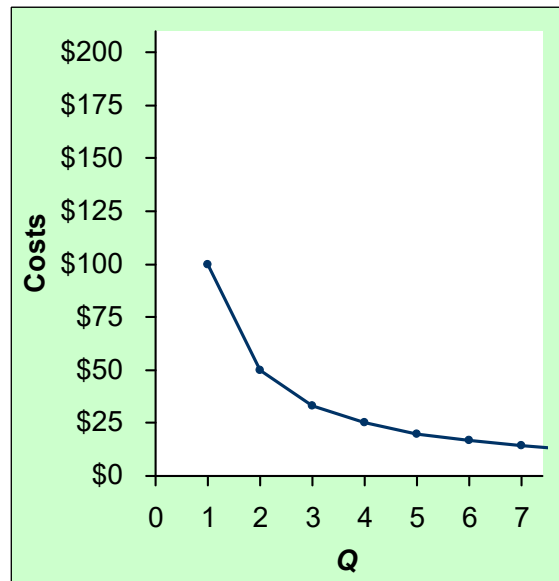


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EXAMPLE 2: Average Fixed Cost

Q	FC	AFC
0	\$100	n/a
1	100	\$100
2	100	50
3	100	33.33
4	100	25
5	100	20
6	100	16.67
7	100	14.29

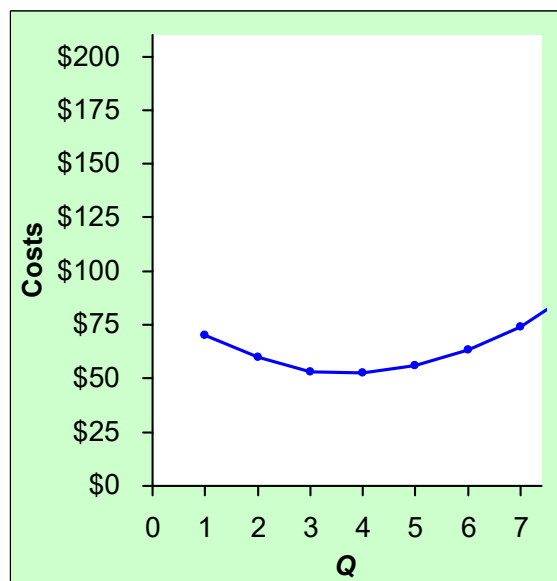


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EXAMPLE 2: Average Variable Cost

Q	VC	AVC
0	\$0	n/a
1	70	\$70
2	120	60
3	160	53.33
4	210	52.50
5	280	56.00
6	380	63.33
7	520	74.29



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EXAMPLE 2: Average Total Cost

Q	TC	ATC	AFC	AVC
0	\$100	n/a	n/a	n/a
1	170	\$170	\$100	\$70
2	220	110	50	60
3	260	86.67	33.33	53.33
4	310	77.50	25	52.50
5	380	76	20	56.00
6	480	80	16.67	63.33
7	620	88.57	14.29	74.29

Average total cost (ATC) equals total cost divided by the quantity of output:

$$ATC = TC/Q$$

Also,

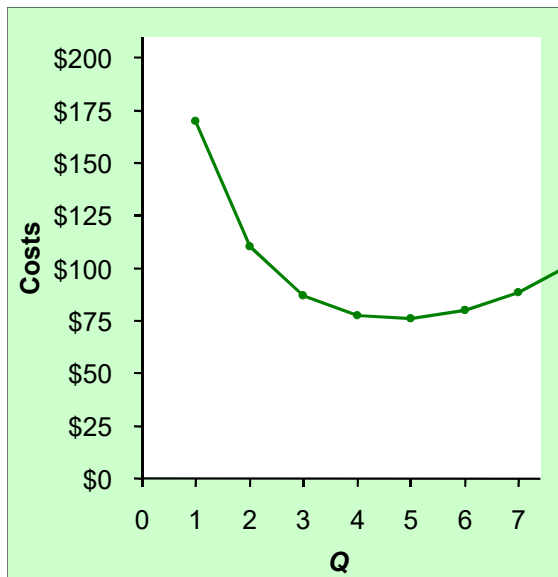
$$ATC = AFC + AVC$$

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EXAMPLE 2: Average Total Cost

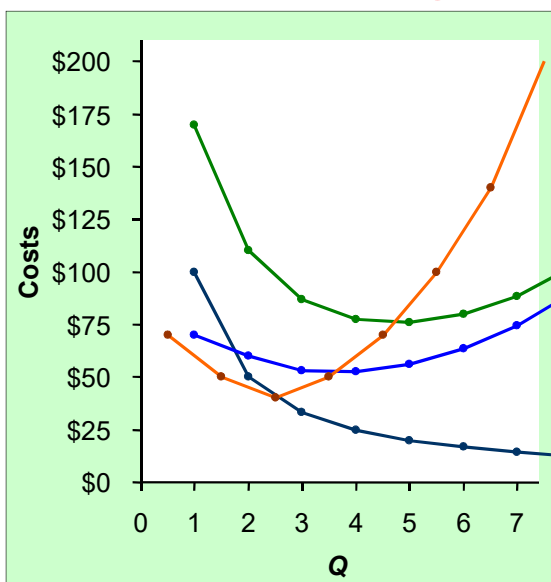
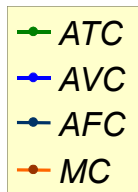
Q	TC	ATC
0	\$100	n/a
1	170	\$170
2	220	110
3	260	86.67
4	310	77.50
5	380	76
6	480	80
7	620	88.57



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EXAMPLE 2: The Various Cost Curves Together



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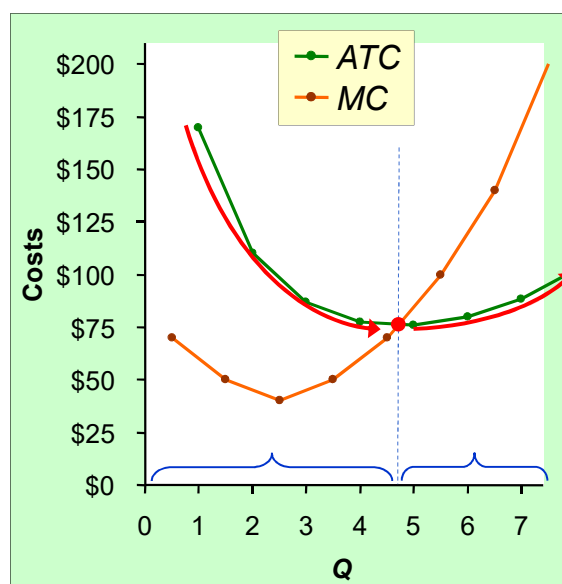


EXAMPLE 2: ATC and MC

When $MC < ATC$,
ATC is falling.

When $MC > ATC$,
ATC is rising.

The MC curve crosses the
ATC curve at
the ATC curve's minimum.



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EXAMPLE 2: Why ATC Is Usually U-Shaped

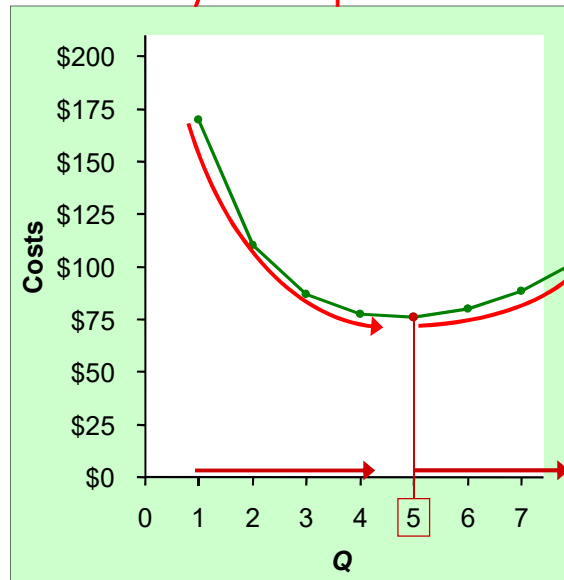
As Q rises:

Initially, falling AFC
pulls ATC down.

Eventually,
rising AVC pulls ATC up.

Efficient scale:

The quantity that
minimizes ATC .



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ACTIVE LEARNING 3

Calculating costs

Fill in the blank spaces of this table.

Q	VC	TC	AFC	AVC	ATC	MC
0		\$50	n/a	n/a	n/a	
1	10			\$10	\$60.00	\$10
2	30	80				
3			16.67	20	36.67	30
4	100	150	12.50		37.50	
5	150			30		
6	210	260	8.33	35	43.33	60

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6.4 Costs in the Short Run & Long Run

- Short run:
Some inputs are fixed (*e.g.*, factories, land).
The costs of these inputs are *FC*.
- Long run:
All inputs are variable
(*e.g.*, firms can build more factories,
or sell existing ones).
- In the long run, *ATC* at any Q is cost per unit using the most efficient mix of inputs for that Q (*e.g.*, the factory size with the lowest *ATC*).



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